

KIRIKKALE, Turkey

WMO: 171350

Lat: 39.8433N

Lon: 33.5181E

Elev: 2454

StdP: 13.44

Time Zone: 3.00 (E03)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.2	18.7	8.1	9.1	17.7	12.0	11.1	22.1	14.4	35.2	12.3	35.3	3.3	80	0.435

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.5	96.4	65.4	93.4	64.8	90.3	64.4	71.2	88.0	69.0	85.3	67.0	83.5	5.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.5	103.3	81.3	63.0	94.4	78.8	60.8	87.1	76.2	36.7	88.4	34.7	85.2	32.9	83.6	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.1	12.1	10.5	DB	11.1	101.9	7.2	3.1	5.9	104.1	1.7	105.9	-2.3	107.6	-7.5	109.8	(4)
			WB	10.1	70.7	7.0	4.8	5.1	74.1	1.0	76.9	-2.9	79.6	-8.0	83.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.1	33.6	38.7	46.9	55.9	62.3	70.3	77.4	77.2	69.0	57.5	46.1	37.2
	DBStd	16.25	8.35	8.61	7.17	6.06	5.54	5.17	4.58	4.98	6.16	6.64	6.43	6.86
	HDD50	1548	509	319	142	15	0	0	0	0	15	149	399	
	HDD65	4392	974	735	563	279	120	12	0	1	31	246	568	863
	CDD50	3769	0	4	44	192	381	608	849	842	571	246	31	1
	CDD65	1140	0	0	1	6	36	170	384	378	153	12	0	0
	CDH74	12989	0	0	9	131	447	1803	4471	4313	1642	173	0	0
	CDH80	5696	0	0	0	17	92	664	2179	2084	628	32	0	0
Wind	WSAvg	3.9	3.1	3.8	4.1	3.9	3.9	4.4	5.4	5.6	3.9	3.2	2.8	2.9
Precipitation	PrecAvg	15.0	1.7	1.3	1.3	1.8	2.1	1.4	0.5	0.3	0.6	1.1	1.2	1.9
	PrecMax	23.6	3.5	3.5	2.5	4.0	3.9	4.4	2.1	1.1	2.4	3.3	4.5	4.1
	PrecMin	9.6	0.1	0.1	0.2	0.2	0.3	0.1	0.0	0.0	0.0	0.0	0.0	0.4
	PrecStd	3.4	1.0	0.9	0.7	1.0	0.9	1.0	0.6	0.3	0.7	0.9	1.0	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.1	64.3	73.3	82.7	86.7	94.1	101.6	100.0	95.3	84.3	68.3	55.6
		MCWB	46.7	49.3	53.3	55.9	59.7	63.4	65.9	64.3	60.5	62.7	52.9	48.0
	2%	DB	50.3	58.1	67.4	77.8	82.7	90.5	97.3	96.3	90.5	78.8	64.4	51.8
		MCWB	44.3	46.8	50.8	55.1	58.7	63.6	66.7	65.2	62.1	58.0	51.4	46.3
	5%	DB	47.1	54.4	63.7	73.6	79.3	87.2	93.7	93.2	86.8	74.6	61.0	49.5
		MCWB	42.6	45.3	49.2	54.5	58.6	62.5	66.0	65.3	61.9	56.9	50.5	45.1
	10%	DB	44.1	50.9	59.8	69.7	75.8	83.7	90.2	90.2	83.3	70.2	57.6	46.9
		MCWB	40.9	43.9	47.6	53.1	58.1	61.6	64.8	65.0	61.2	55.3	48.5	43.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.3	50.3	55.7	60.0	64.4	70.0	75.5	73.7	68.3	66.2	55.5	50.0
		MCDB	53.0	60.4	69.0	73.6	77.3	86.7	93.9	88.8	84.5	81.6	65.2	52.7
	2%	WB	45.0	47.9	52.4	57.6	61.9	66.9	72.0	71.1	66.3	60.3	53.0	47.4
		MCDB	49.2	55.9	64.0	71.6	75.6	81.9	89.2	87.5	83.4	73.2	61.4	50.9
	5%	WB	43.1	46.2	50.2	55.8	60.3	64.7	69.4	69.0	64.3	58.2	51.1	45.6
		MCDB	46.7	52.9	60.9	70.0	74.3	80.9	86.0	85.0	80.0	69.5	58.7	48.7
	10%	WB	41.2	44.3	48.3	54.0	59.0	63.1	67.0	67.0	62.2	56.7	49.4	43.8
		MCDB	44.0	50.4	58.4	67.2	72.8	79.3	83.6	83.9	77.5	67.4	56.4	46.6
Mean Daily Temperature Range	5% DB	MDBR	11.1	15.5	19.4	22.2	21.9	23.2	24.5	24.0	24.5	21.4	19.9	11.1
		MCDBR	15.3	21.5	26.4	28.9	27.4	28.0	28.4	27.5	29.8	28.1	24.1	14.3
		MCWBR	10.0	12.1	13.8	12.4	10.3	9.1	9.6	8.5	11.5	13.3	13.6	9.6
	5% WB	MCDBR	14.1	18.6	23.0	25.5	24.4	24.6	25.7	24.0	26.0	23.4	21.0	12.4
		MCWBR	9.7	11.0	12.7	12.2	10.2	10.0	10.9	10.0	12.0	11.7	12.4	8.9
Clear-Sky Solar Irradiance	taub	0.314	0.343	0.387	0.428	0.404	0.381	0.380	0.378	0.356	0.359	0.325	0.315	
	taud	2.356	2.293	2.189	2.104	2.244	2.334	2.336	2.318	2.379	2.358	2.422	2.411	
	Ebn at Noon	268	275	273	269	278	284	283	280	281	266	263	259	
	Edh at Noon	28	34	42	49	43	40	39	39	34	31	26	24	
All-Sky Solar Radiation	RadAvg	595	924	1290	1652	1962	2247	2379	2126	1676	1145	787	528	
	RadStd	74	137	112	152	142	170	95	78	110	82	74	68	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)		+0.90	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page